

massi: Towards Human-AI Scientific Linked Data Co-Creation



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Motivation

Standard metadata approaches often fail to capture **tacit knowledge**, e.g. expert intuition, situational context, impeding data reuse [1].

We believe **hybrid intelligence** lens can augment knowledge elicitation. Through human-AI cooperative dialogue, experts' experiential awareness can be embedded into structured linked data, preserving the broader **scientific context for future reuse**.

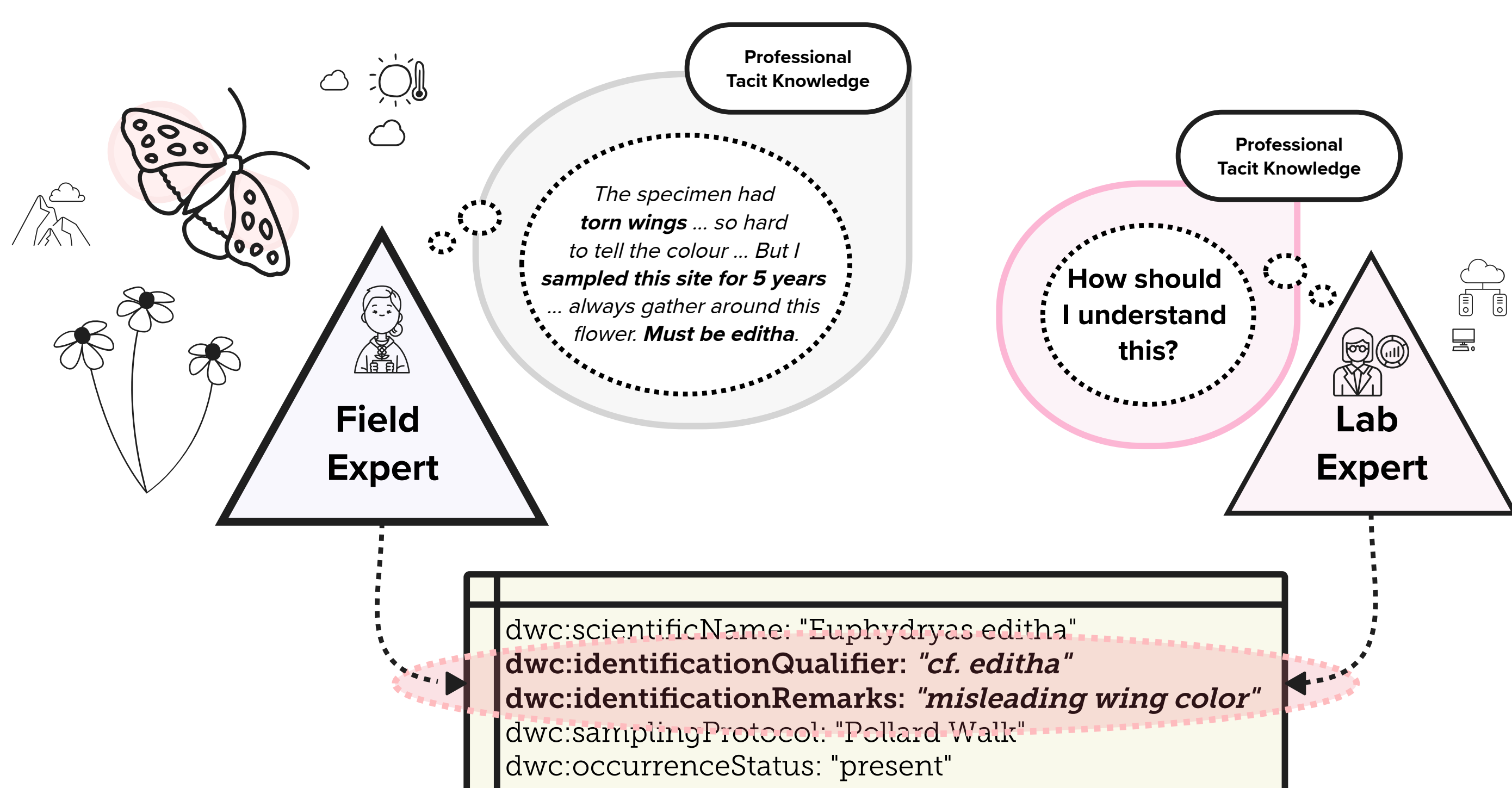


Figure 2: lost tacit knowledge complicates data reuse.

Hybrid-Intelligence Lens

Centering Human-AI cooperation in a scientific data work [2]:

- **Collaborative:** join human and artificial intelligence
- **Adaptive:** reflect and use complementary skills
- **Responsible:** align with ethical principles
- **Explainable:** mutually transparent intentions



Cooperation over automation

Research Methods

Research bridge for technical and social:

- **Design Science Research**
 - multi-agent systems for semantic web
- **Socio-Technical Systems**
 - multiple practitioners' and systems
- **More-than-Human Interaction**
 - agentic human-ai and human-data teamwork
- **Collective Knowledge Elicitation**
 - knowledge dialogue instead of extraction

Open Challenges

Approaching data reuse as an entangled practice [3]:

- Reflecting the scientists' needs?
- Cooperation with multiple scientific authors?
- Supporting relationship with data stewards?
- Handling (linked) data infrastructures?
- Using community standards and ontologies?
- Fitness to diverse scientific work settings?

References

- [1] Thanos, C. Research Data Reusability: Conceptual Foundations, Barriers and Enabling Technologies. *Publications* 5, 2 (2017).
- [2] De Boer, V. & Stork, L. Hybrid Intelligence for Digital Humanities. in *Frontiers in Artificial Intelligence and Applications* (eds Lorig, F. et al.) (IOS Press, 2024).
- [3] Thylstrup, N. B., Hansen, K. B., Flyverbom, M. & Amoores, L. Politics of data reuse in machine learning systems: Theorizing reuse entanglements. *Big Data & Society* 9, 20539517221139785 (2022).

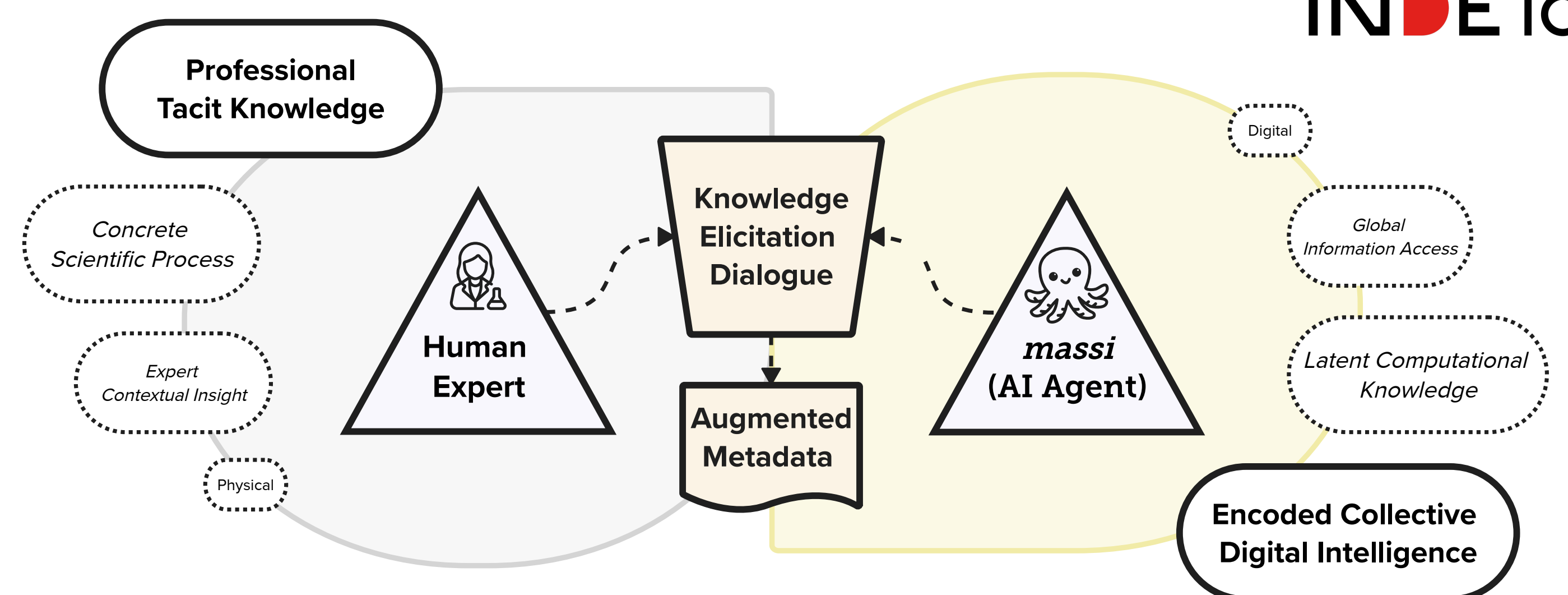


Figure 1: contributions to knowledge elicitation dialogue between a human data creator and massi.

Research Questions

- How to **evaluate tacit knowledge** representation in scientific datasets?
- How to **effectively coordinate human-AI** metadata co-creation workflows?
- How to **co-design Agentic AI systems** with scientific communities?

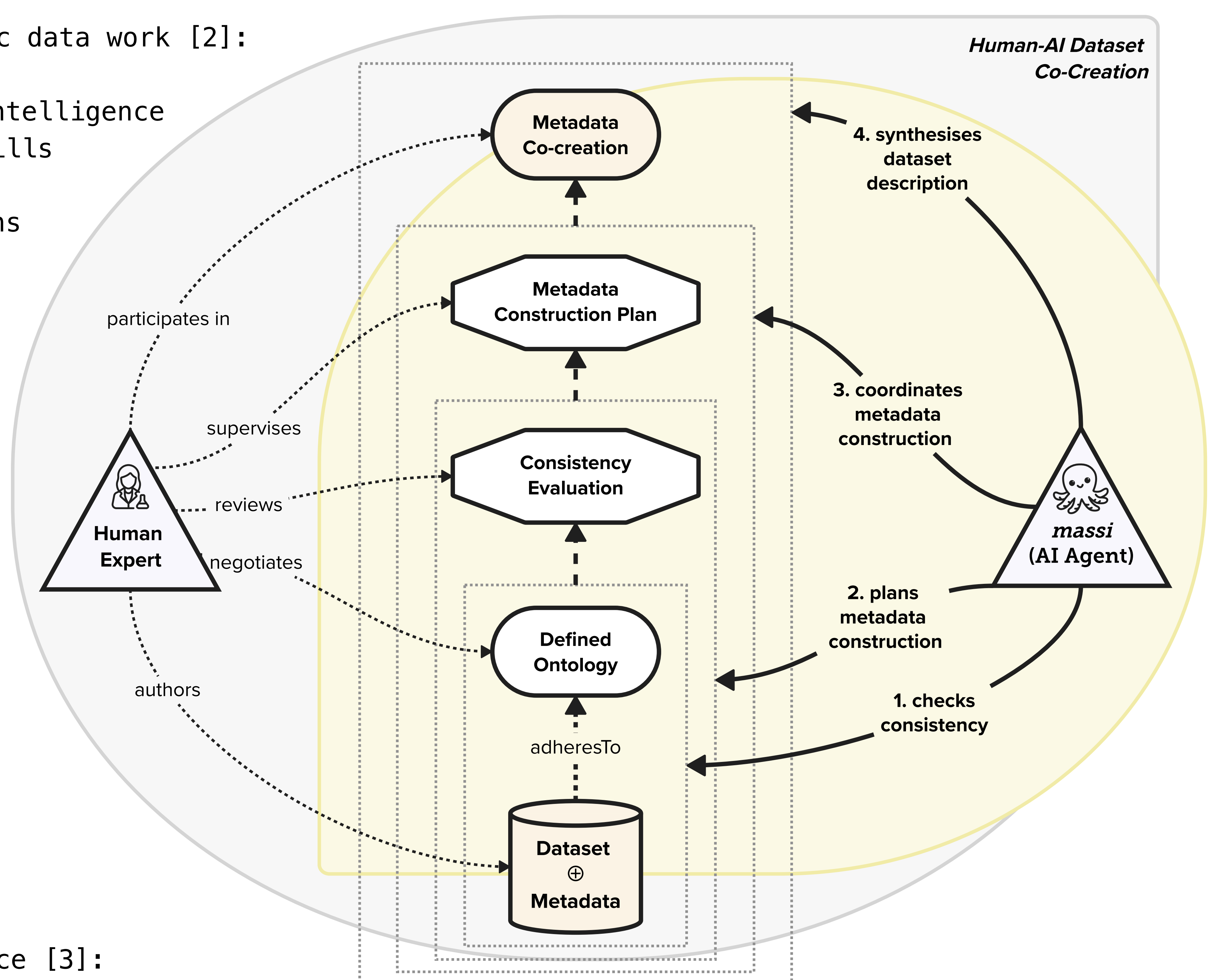


Figure 3: cooperative metadata construction process with a human data creator and a massi agent.

Takeaways

- massi aims to **cooperatively elicit expert tacit knowledge** during the creation of scientific linked data.
- Do you **publish data? Let's talk!**